CFD_HW_7

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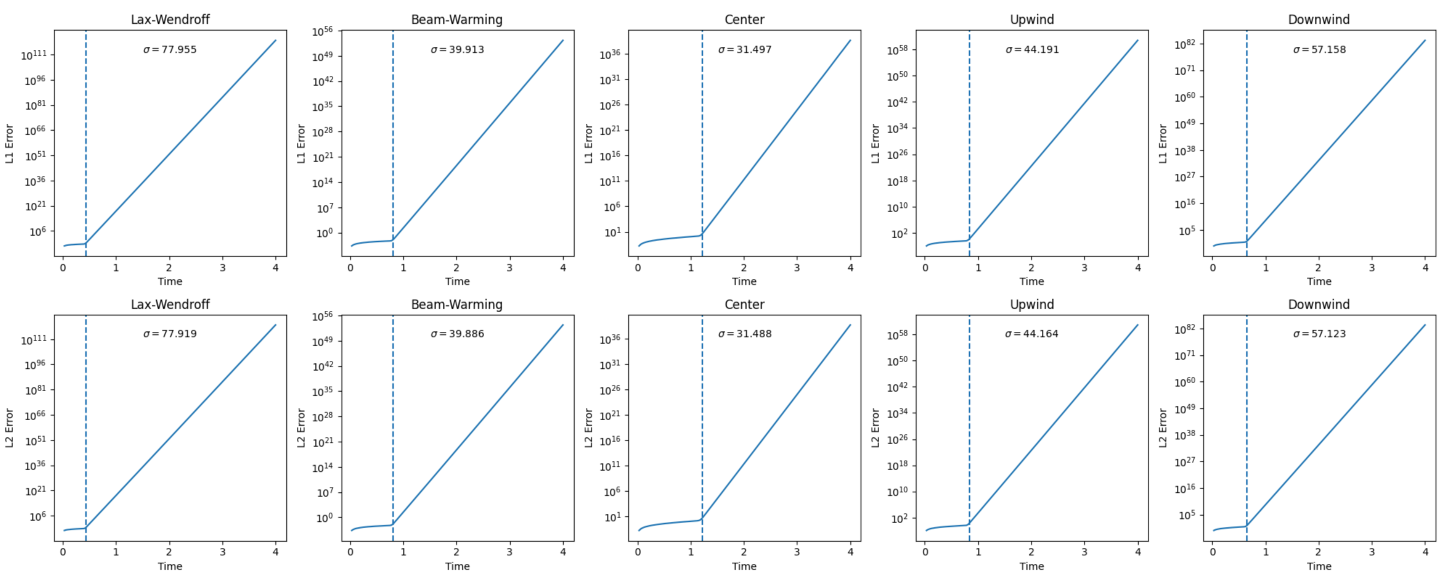
19. Error Slopes

(a) & (b)

The answer for (a) and (b) are all below:



The slope marks I use " σ " has reason which will be explained in (c).

(c) Predict the slope

Interpretation of the Error Growth Rate σ

1. Exponential error growth

For numerical solutions of the linear advection equation, the error often evolves as:

$$e(t) = u_{\text{num}}(t) - u_{\text{exact}}(t)$$

In unstable or marginally stable regimes, the error exhibits exponential behavior:

$$e(t) \sim e^{\sigma t}$$

Taking the logarithm:

$$\log |e(t)| \sim \sigma t$$

Thus, the slope of the log-error curve is the growth rate σ.

2. Explaination

Using von Neumann analysis for the errors:

$$e_j^n = e^{\sigma t} e^{ikj\Delta x}$$

And in L1 or L2 :

$$L1(t) = \frac{1}{N} \sum_j |e_j(t)| \approx \frac{1}{N} |e^{\sigma t}| \sum_j |e^{ikj\Delta x}| = C_1 e^{\sigma t}$$

$$L2(t) = L2(t) = \sqrt{\frac{1}{N} \sum_j e_j(t)^2} \approx |e^{\sigma t}| \sqrt{\frac{1}{N} \sum_j e^{ikj\Delta x^2}} = C_2 e^{\sigma t}$$

And in the plots, the y-axis is log mod:

$$\log L1(t) = \sigma t + \log C1$$

$$\log L2(t) = \sigma t + \log C2$$

Therefore:

$$\frac{d}{dt} \log(L1) = \frac{d}{dt} \log(L2) = \sigma$$

σ and CFL (C = 2.5) for different schemes

For linear advection with periodic boundary conditions:

$$u(x, t = 0) = \sin(\pi x) + 0.6 \sin(3\pi x), \quad x \in [0, 2]$$

the discrete error evolution follows:

$$\sigma(k) = \frac{1}{\Delta t} \ln |G(C, k\Delta x)|$$

where we consider G by von Neumann analysis:

$$G(C, k\Delta x) = |e^{\sigma t}|$$

with:

$$C = \frac{a\Delta t}{\Delta x} = 2.5$$

Grid resolution:

$$\Delta x = \frac{2}{N} = \frac{2}{160} = 0.0125$$

So:

- $k_1 = \pi \Rightarrow k_1 \Delta x \approx 0.0393$
- $k_2 = 3\pi \Rightarrow k_2 \Delta x \approx 0.1180$

Dominant growth rate:

$$\sigma = \max(\sigma_{k_1}, \sigma_{k_2})$$

1. Upwind scheme

Small-k approximation:

$$G \approx 1 - C(1 - e^{-ik\Delta x})$$

Estimated growth:

- for k_1: weak growth
- for k_2: stronger growth

At $C = 2.5$:

$$\sigma \approx \frac{1}{\Delta t} \ln(1.8 \sim 2.2) \approx 40 \sim 70$$

2. Lax–Wendroff

$$G = 1 - iC \sin(k\Delta x) + C^2(\cos(k\Delta x) - 1)$$

Small-angle approximation:

- dispersive amplification dominates when $C > 1$

Estimated:

$$|G| \approx 1.3 \sim 1.6$$

$$\sigma \approx \frac{1}{\Delta t} \ln(1.3 \sim 1.6) \approx 15 \sim 40$$

3. Beam–Warming

Second-order upwind bias → stronger CFL sensitivity

$$G = 1 - C(1 - e^{-ik\Delta x}) - \frac{C(1-C)}{2}(1 - e^{-ik\Delta x})^2$$

At $C = 2.5$:

$$|G| \approx 1.5 \sim 2.0$$

Estimated:

$$\sigma \approx 25 \sim 60$$

4. Centered scheme

$$G = 1 - iC \sin(k\Delta x)$$

Magnitude:

$$|G| = \sqrt{1 + C^2 \sin^2(k\Delta x)}$$

For small $k\Delta x$:

$$|G| \approx 1 + \frac{1}{2} C^2 (k\Delta x)^2$$

Estimated:

$$|G| \approx 1.01 \sim 1.03$$

$$\sigma \approx 5 \sim 15$$

5. Downwind scheme

$$G = 1 + C(1 - e^{-ik\Delta x})$$

Strong amplification:

$$|G| \approx 2.5 \sim 3.5$$

Estimated:

$$\sigma \approx 50 \sim 90$$

Summary (C = 2.5)

Scheme	Estimated σ range
Upwind	40 ~ 70
Lax–Wendroff	15 ~ 40
Beam–Warming	25 ~ 60
Center	5 ~ 15
Downwind	50 ~ 90

20. Error Slopes

(a) The effect of Leading-edge trunction error

1. Upwind Scheme

The modified equation is :

$$u_t + au_x = \frac{a\Delta x}{2}(1 - C)u_{xx} + O(\Delta x^2)$$

The Leading-edge Trunction Error is :

$$\mathrm{LTE} \sim (1 - C)\Delta x$$

Properties:

- First-order accurate
- Dominated by numerical diffusion
- Error decreases as $C \to 1$

2. Lax–Wendroff Scheme

The modified equation is :

$$u_t + au_x = \frac{a(\Delta x)^2}{6}(C^2 - 1)u_{xxx} + \frac{a(\Delta x)^3}{8}C(1 - C^2)u_{xxxx} + \cdots$$

The Leading-edge Trunction Error is :

$$\mathrm{LTE} \sim (C^2 - 1)(\Delta x)^2$$

Properties:

- Second-order accurate
- Dominated by dispersive error
- Error decreases significantly as $C \to 1$

3. Scheme B

The modified equation is :

$$u_t + au_x = \frac{a\Delta x^2}{6}(1-C)(2-C)u_{xxx} + \cdots$$

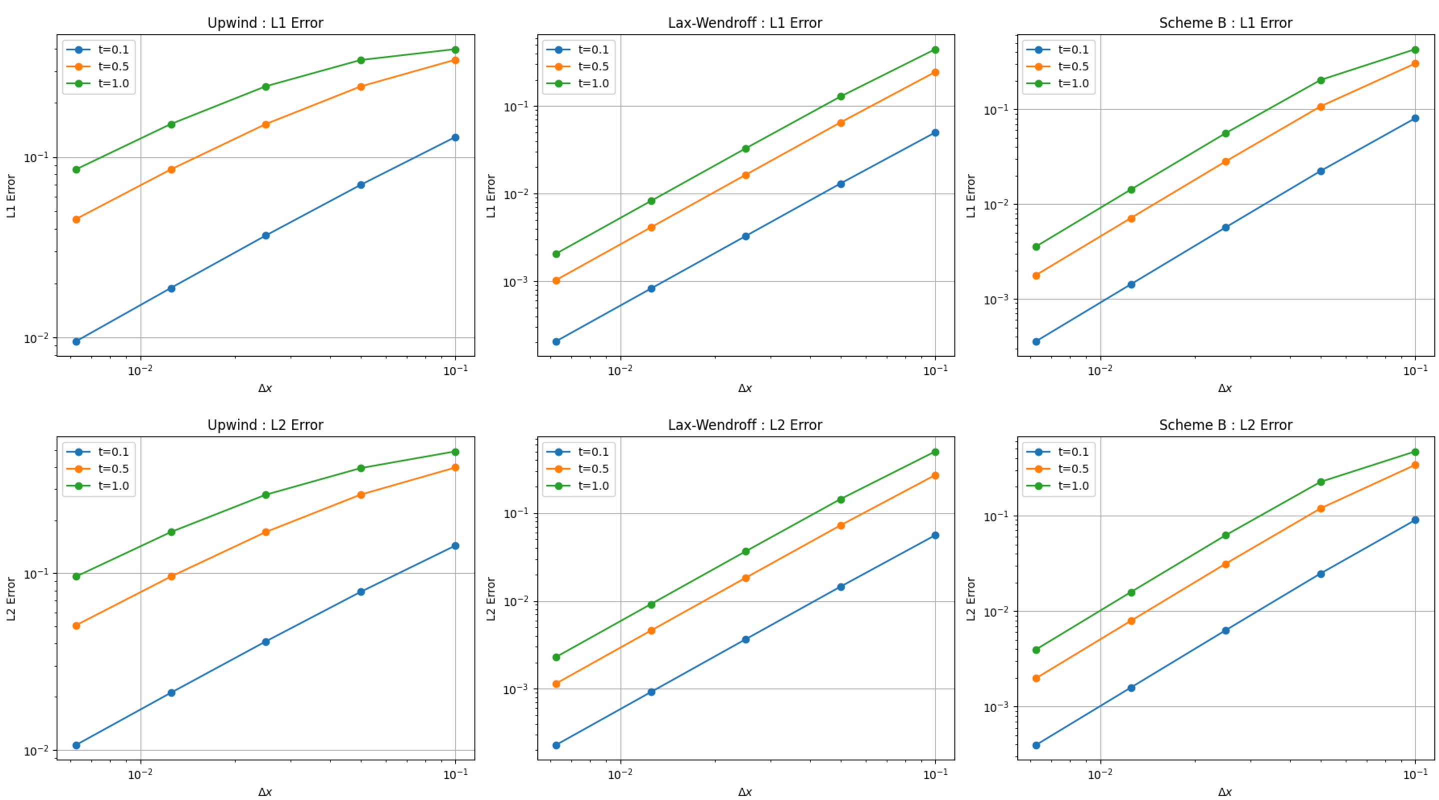
The Leading-edge Trunction Error is :

$$\text{LTE} \sim (1-C)(2-C)(\Delta x)^2$$

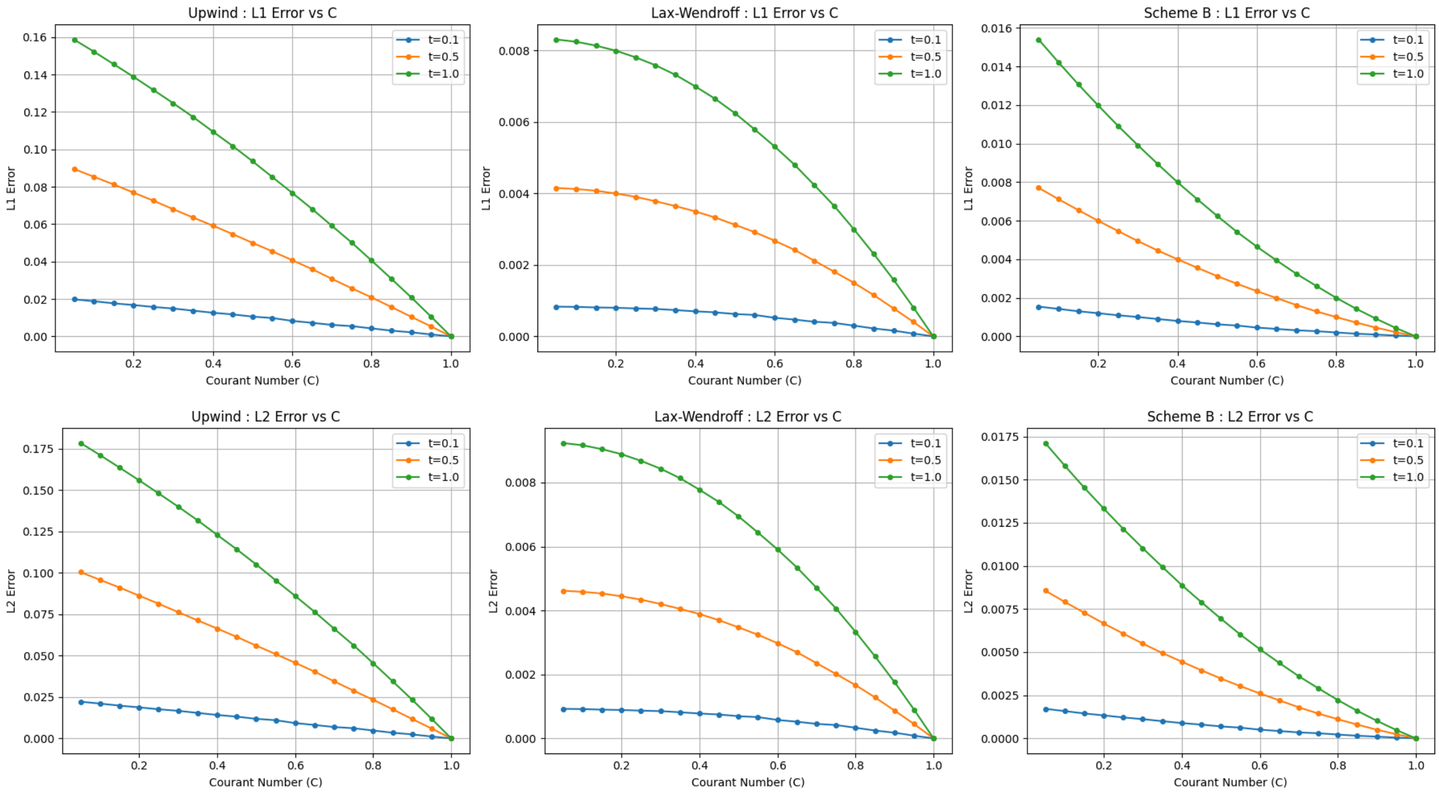
Properties:

- Second-order accurate
- Mainly dispersive
- Error decreases significantly as $C \to 1$ and $C \to 2$
- More stable and less oscillatory than Lax–Wendroff

(b) Varied by Δx



(c) Varied by C



(d) Prediction

1. For (b), The orde of Δx represent the slope of the plots.

Because the Courant number C is fixed at 0.1, the C-dependent terms become constants. The truncation error is entirely governed by the spatial step size, Δx.

- **Upwind Scheme:** The error is proportional to Δx^1.
  - **Prediction:** On a log-log plot, the L1 and L2 error curves will show a linear decrease with a **slope of 1**, demonstrating first-order spatial accuracy.
- **Lax-Wendroff Scheme:** The error is proportional to Δx^2.
  - **Prediction:** On a log-log plot, the error curves will decline more steeply with a **slope of 2**, demonstrating second-order spatial accuracy.
- **Scheme B:** The error is also proportional to Δx^2.

- **Prediction:** On a log-log plot, the curves will also exhibit a **slope of 2**. However, because the coefficient $(1-C)(2-C)$ differs from the Lax-Wendroff coefficient $(1-C^2)$ when $C = 0.1$, the Scheme B curve will be parallel to the Lax-Wendroff curve but with a different vertical intercept.

2. For (c), C represent the shape of the plots.

Because the grid size N (and therefore Δx) is fixed, the spatial component of the error term acts as a constant. The error variation is entirely driven by how the C-dependent coefficient changes as $0 < C \leq 1.0$.

- **Upwind Scheme:** The error coefficient is proportional to $(1-C)$.
  - **Prediction:** On a standard linear plot, the error will show a **linear decrease** as C increases from 0 toward 1.
- **Lax-Wendroff Scheme:** The error coefficient is proportional to $(1-C^2)$.
  - **Prediction:** The error will exhibit a **parabolic (quadratic) decay** as C increases.
- **Scheme B:** The error coefficient is proportional to $(1-C)(2-C)$.
  - **Prediction:** The error will also trace a **quadratic decay curve** over the interval $0 < C \leq 1.0$.

3. For (b) and (c), the several representative times shows how the Leading-edge Trunction Error influence the L1 and L2 error.

Short-term errors are governed by the scheme's formal "order of accuracy," but long-term results are ruined by cumulative dissipation, dispersion, and boundary interference.

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